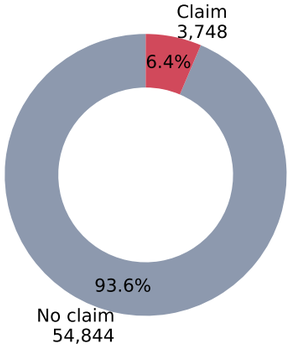
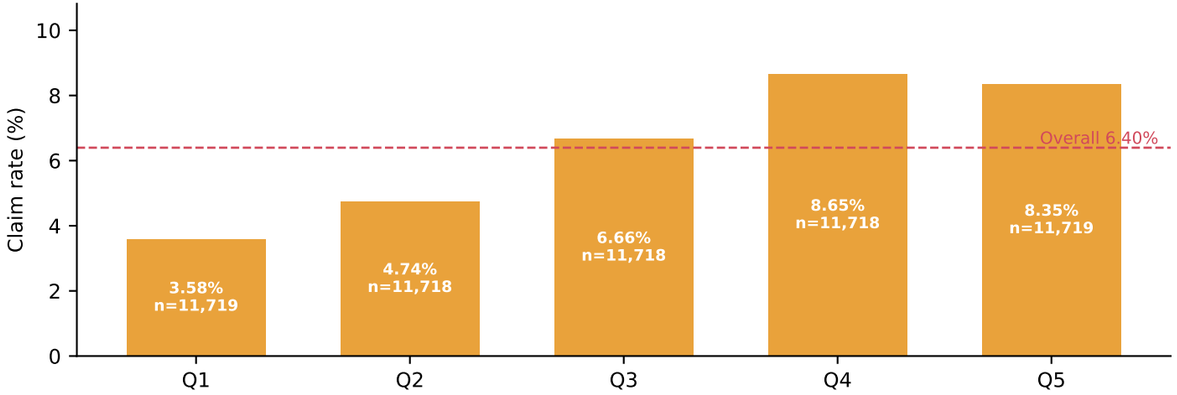
SafeDrive Insurance Ltd. — Claims Risk Dashboard

58,592	3,748	6.40%	761	1.76
Total Policies	**Total Claims**	**Claim Rate**	**High-Risk Customers**	**Average NCAP Rating**
all policies in dataset	is_claim = 1	claims / policies	High + Very High, scored test set	scale 0-5, all policies

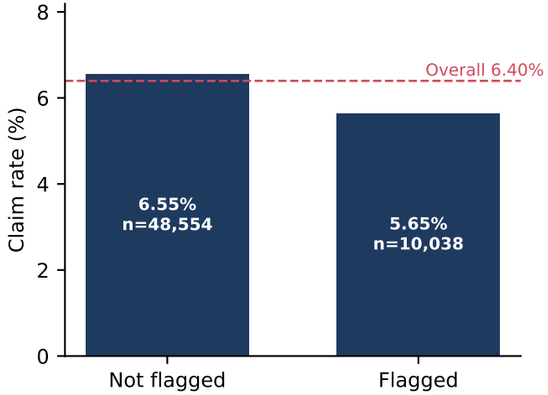
Claim Distribution



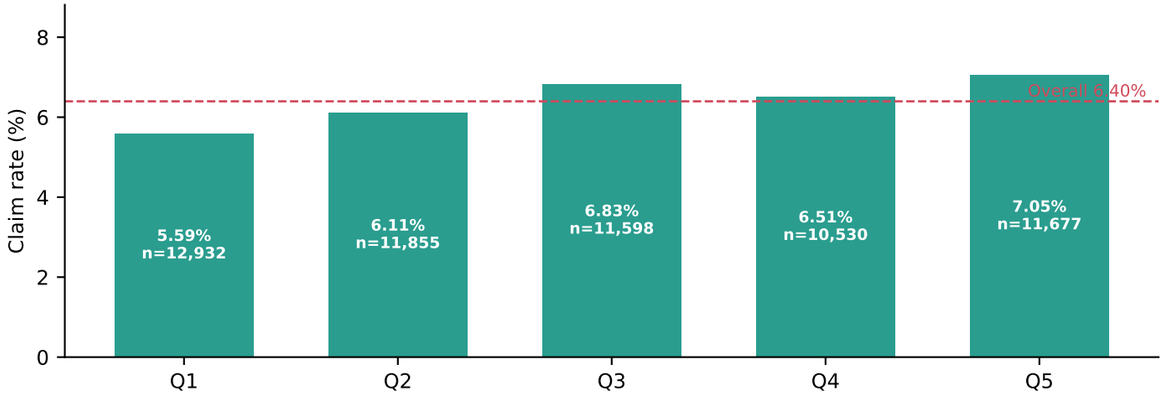
Claim Rate by Policy Tenure Quintile (Q1 shortest → Q5 longest)



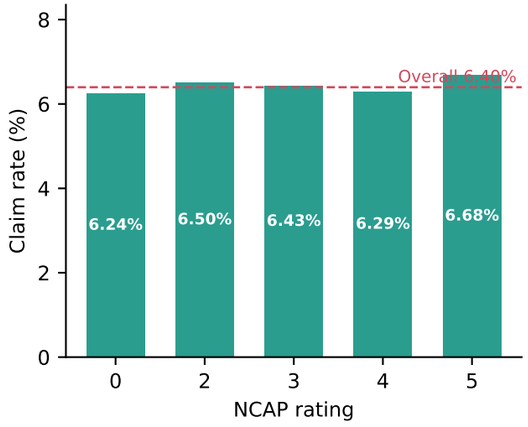
Driver Segment: High-Risk Flag



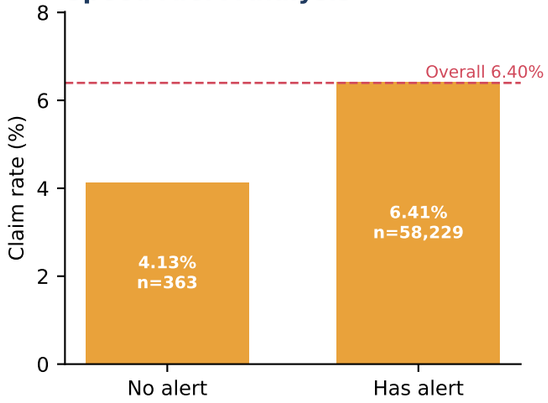
Driver Risk Segmentation: Policyholder Age Quintile (normalised age; Q1 youngest)



NCAP Rating Analysis



Speed Alert Analysis



Notes: red dashed line = overall claim rate. NCAP rating 1 does not occur in the data. Only 363 policies lack a speed alert (15 claims), so that comparison is statistically weak. High-Risk Driver Flag = 1 if policyholder age <= 25th percentile (0.365) AND NCAP rating <= 2, else 0.

SafeDrive Insurance Ltd. — Model Output & Predicted Risk

11,719
Scored Test Policies
20% hold-out

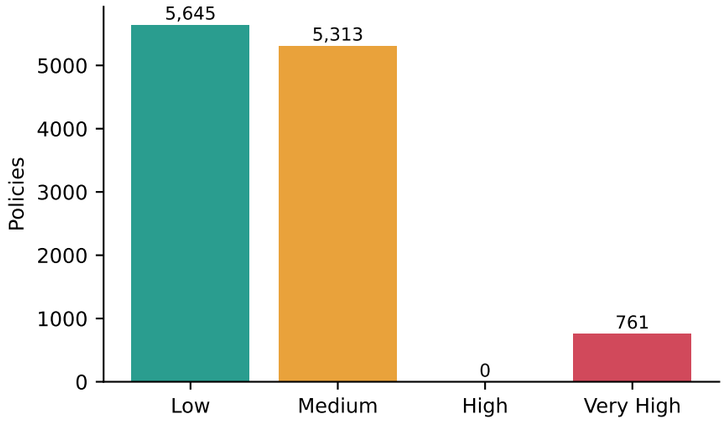
750
Test Claims
actual outcomes

6.40%
Test Claim Rate
actual

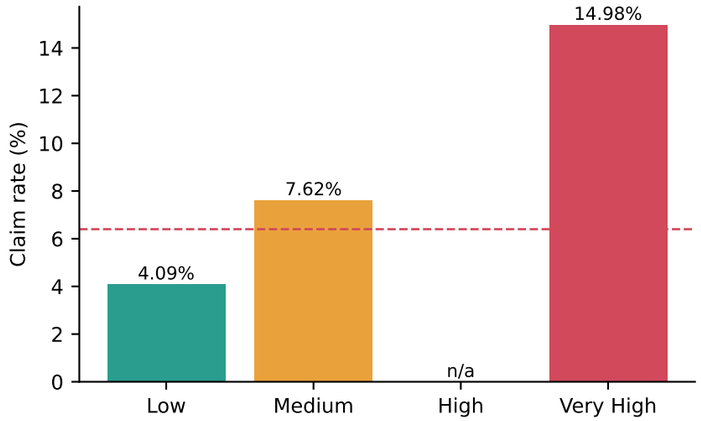
761
Very High Risk
High band is empty (see note)

0.646
Test ROC-AUC
5-fold CV 0.638 ± 0.013

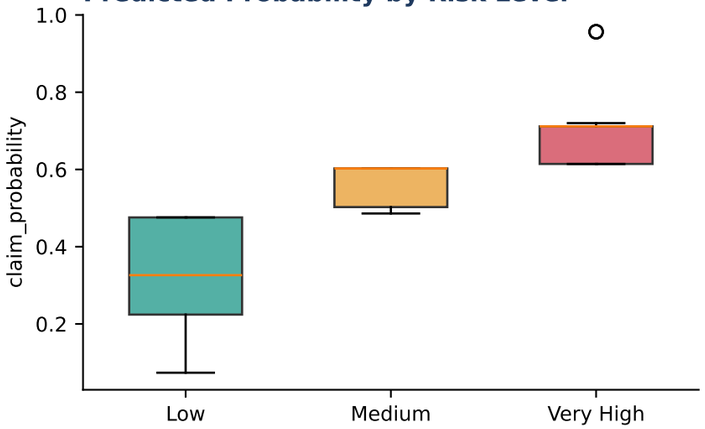
Policies by Predicted Risk Level



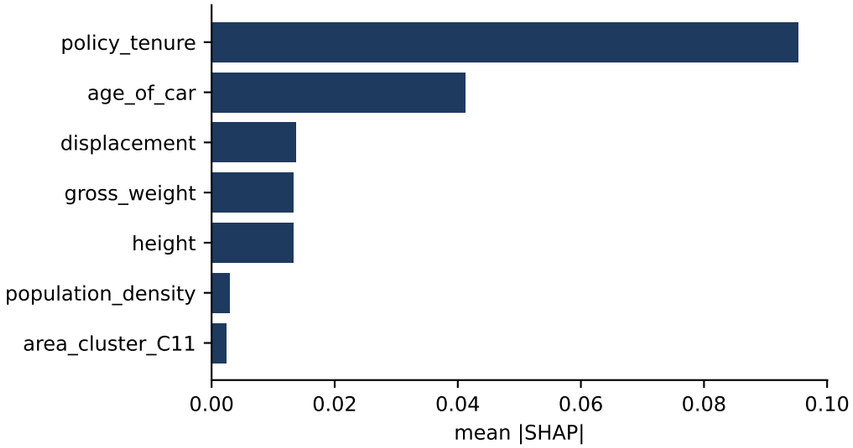
Actual Claim Rate by Risk Level



Predicted Probability by Risk Level



Top Drivers of Predicted Risk (mean |SHAP|)



Model Comparison (test set)

Model	ROC-AUC	Precision	Recall	F1	CV AUC
Logistic Regression	0.585	0.080	0.548	0.140	0.610
Decision Tree (selected)	0.646	0.089	0.653	0.157	0.638

Reading the model output

- Claim rate rises with predicted risk: Low 4.09% → Medium 7.62% → Very High 14.98% (monotonic across the bands that exist).
- Very High is ~3.7× the Low band and 2.3× the overall rate.
- Policy tenure and car age drive most of the tree's predictions.
- Modest discrimination (AUC 0.65): use as a screening aid, not an automatic decision.

Note: the tree produces only 16 distinct probabilities, so the 75th and 90th percentile cut-points coincide (0.6026) and the "High" band is empty; High-risk count therefore equals the Very High count (761). Model is not approved for production.